Use case Diagram

kst@ucsc.cmb.ac.lk
tss@ucsc.cmb.ac.lk

What is Use Case Diagram?

A Use Case Diagram is a type of UML (Unified Modeling Language) diagram used to model the functional behavior of a system from the user's perspective.

- It shows who interacts with the system (actors) and what the system does (use cases).
- It is categorized under behavioral diagrams in UML because it represents the dynamic behavior of the system.

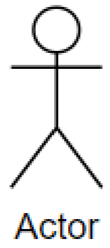
Purpose of Use Case Diagram

- To capture system requirements.
- To visualize user interactions with the system.
- To identify and clarify system functionalities.
- To serve as a communication tool between stakeholders (business analysts, developers, clients, testers).

Notation Description

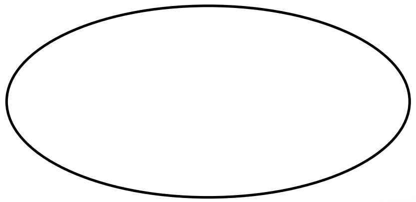
Actor:

➢ Represents a user or system that interact with your system.



Use Case:

➢ An oval shape showing a function the system performs.



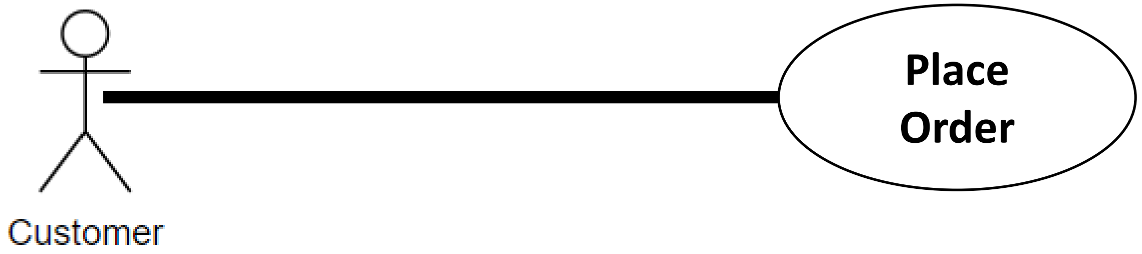
Notation Description

System Boundary:

➢ Defines the scope of the system.

Association:

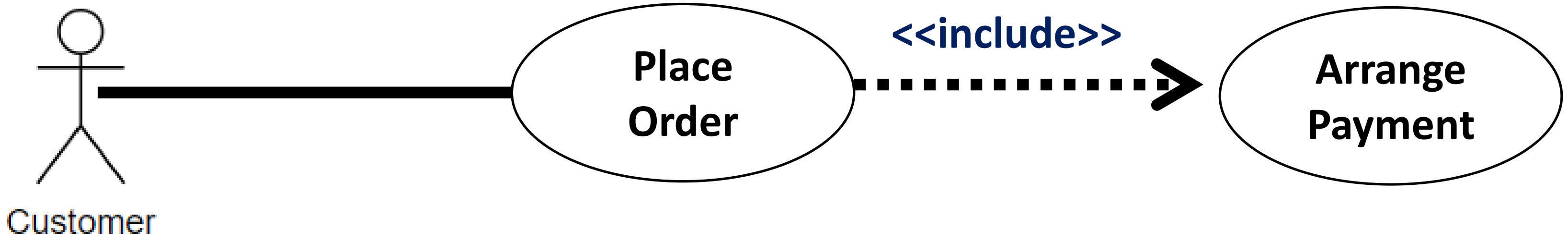
➢ Shows interaction between actor and use case.



Notation Description

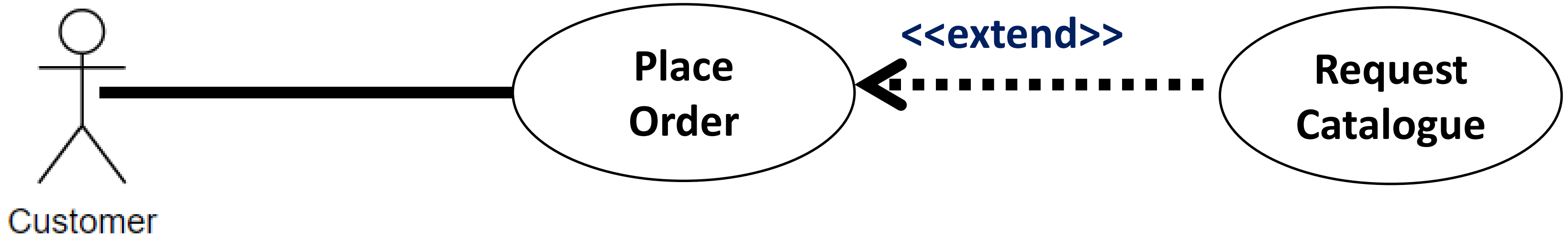
Include:

- Represents mandatory inclusion of another use case.



Extend:

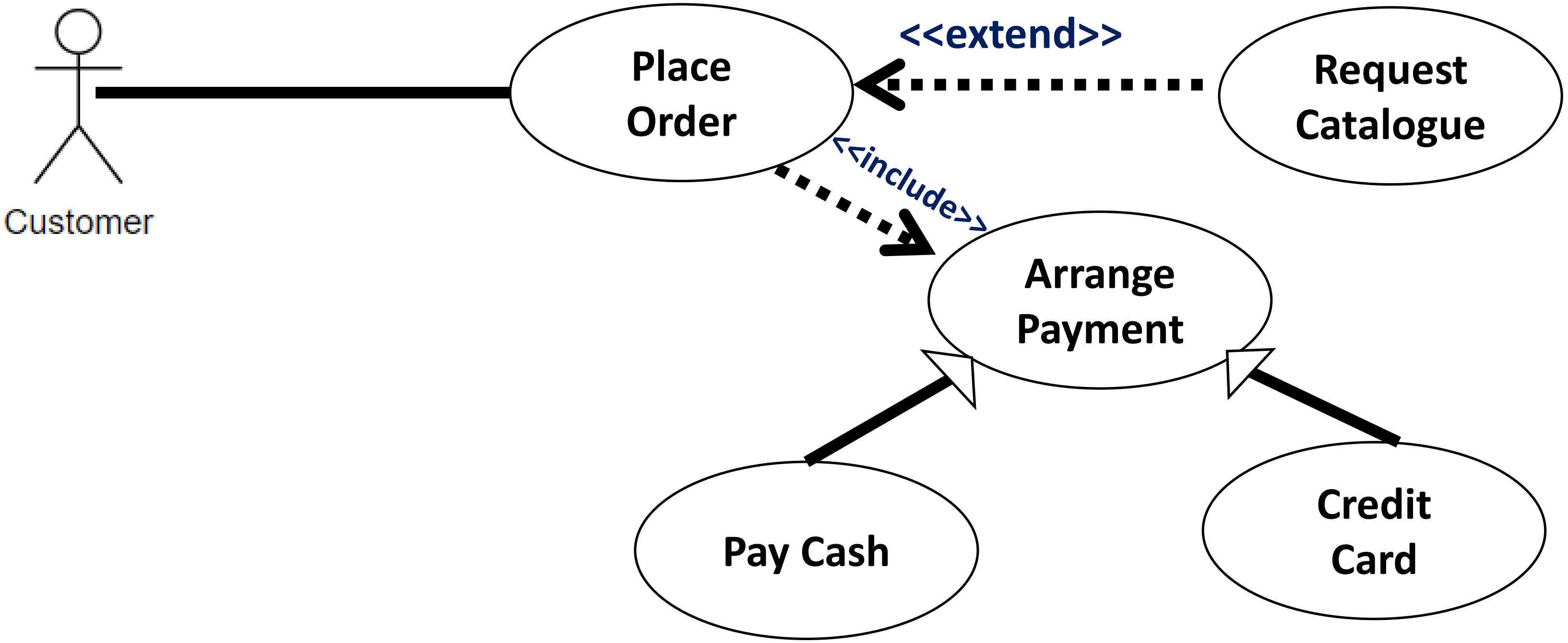
- Represents an optional behavior that extends a base use .



Notation Description

Generalization:

- Used when actors or use cases inherit behavior from others.



Quiz - 01

Haritha is a student at the UCSC. He logs in to the library portal to search for a book and reserve it for borrowing. Who is Haritha in the system, and what actions (use cases) can he perform?

Answer

Haritha is an actor(Student).

He can perform actions like:

- Login
- Search Book
- Reserve Book

Quiz - 02

An online examination system allows students to log in, view available exams, and submit answers. Once the exam is submitted, the system automatically calculates the score. Examiners can create exams and view student scores.

- Identify the actors and use cases involved.
- What are the actions unique to each actor?

Answer

I. Actors: Student, Examiner

II. Student:

✍ Login

✍ View Available Exams

✍ Attempt Exam

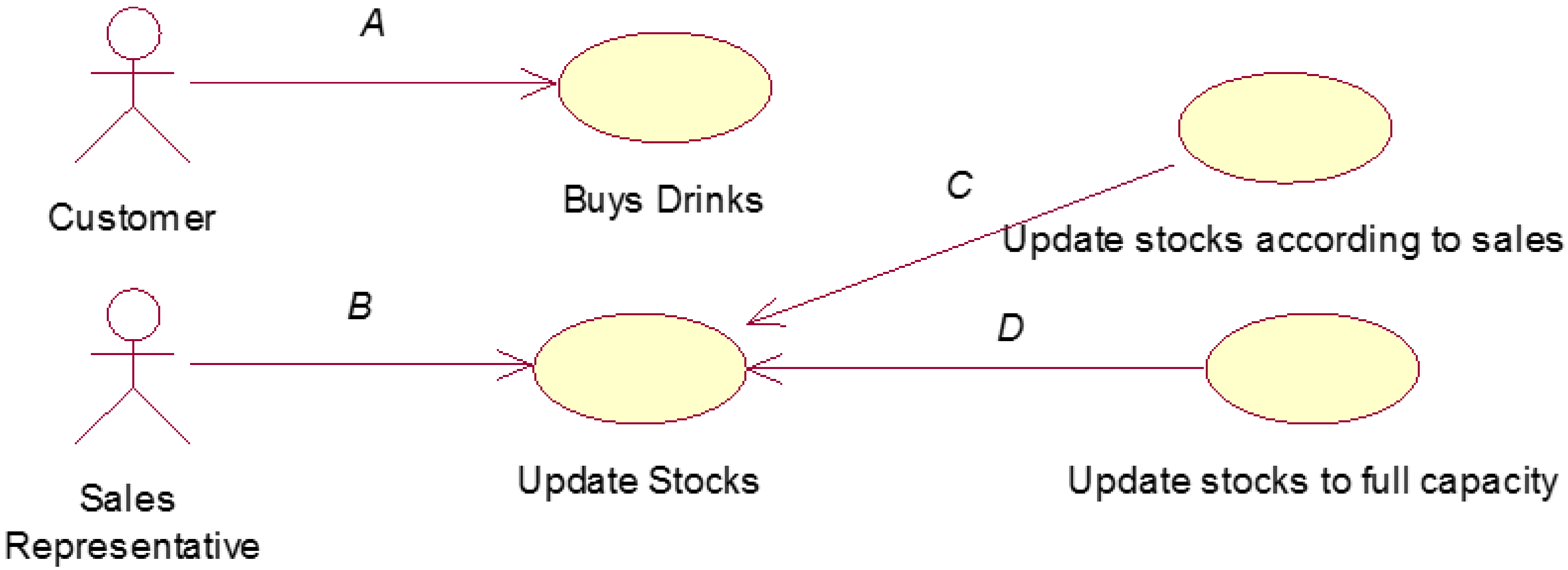
✍ Submit Answers

Examiner

✍ Create Exam

✍ View Student Scores

3. Consider the following use case diagram.

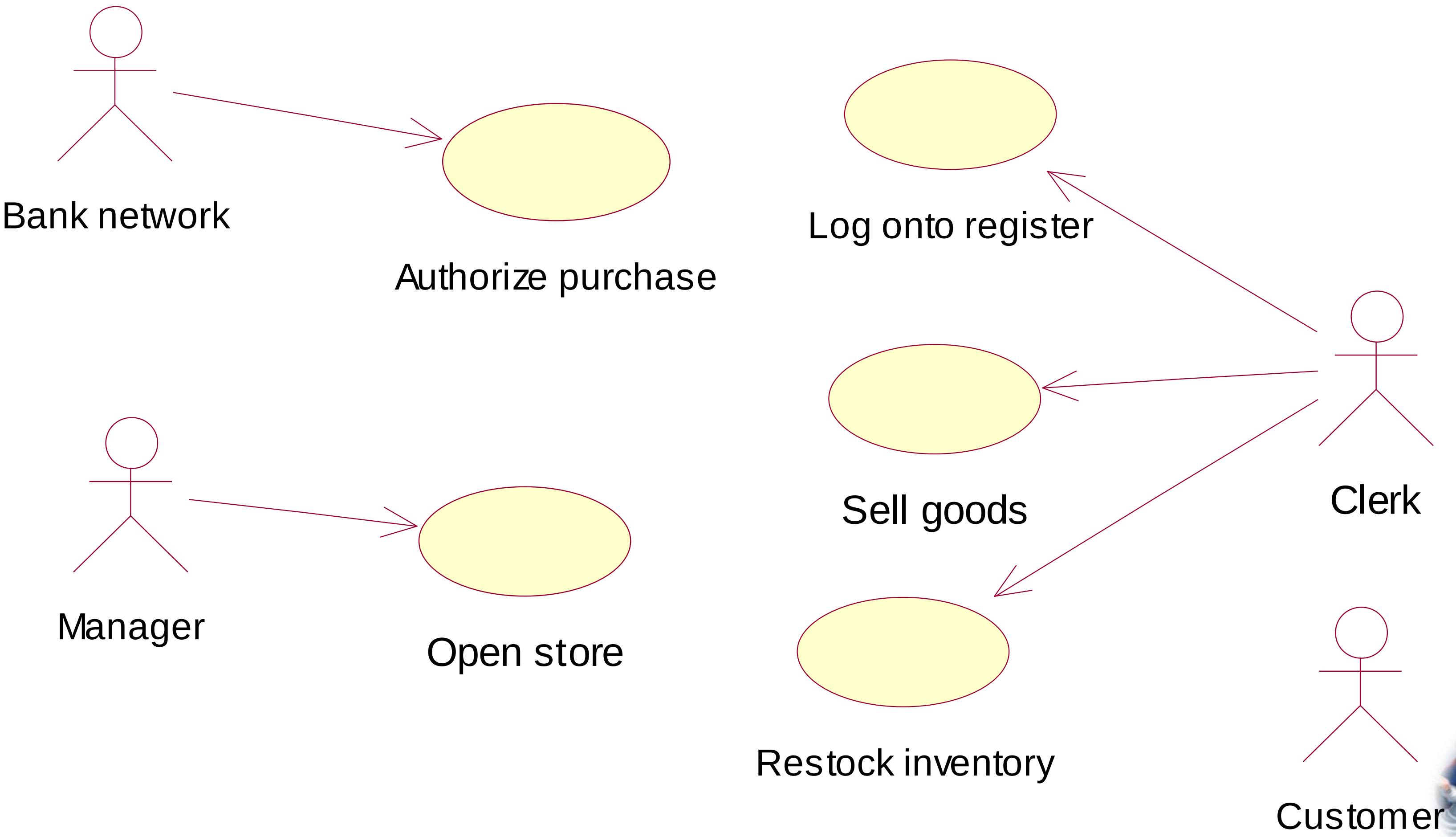


3. Which of the following gives the correct matching(s) for A-D?

- (a) A-Association, B-Association, C-Include, D- Include
- (b) A-Association, B-Association, C-Extend, D- Include
- (c) A-Association, B-Association, C-Extend, D- Extend
- (d) A-Communicate, B-Communicate, C-Inheritance, D- Inheritance
- (e) A-Communicate, B-Communicate, C-Generalization, D- Generalization

Answer : C

4. Consider the following use case diagram drawn for a particular scenario.



4. Which of the following gives the correct matching(s) for A-D?

- (a) Clerk, Manager, Customer
- (b) Clerk, Manager
- (c) Clerk, Manager, Bank network
- (d) Clerk, Manager, Bank network, Customer
- (e) Manager, Bank network, Customer

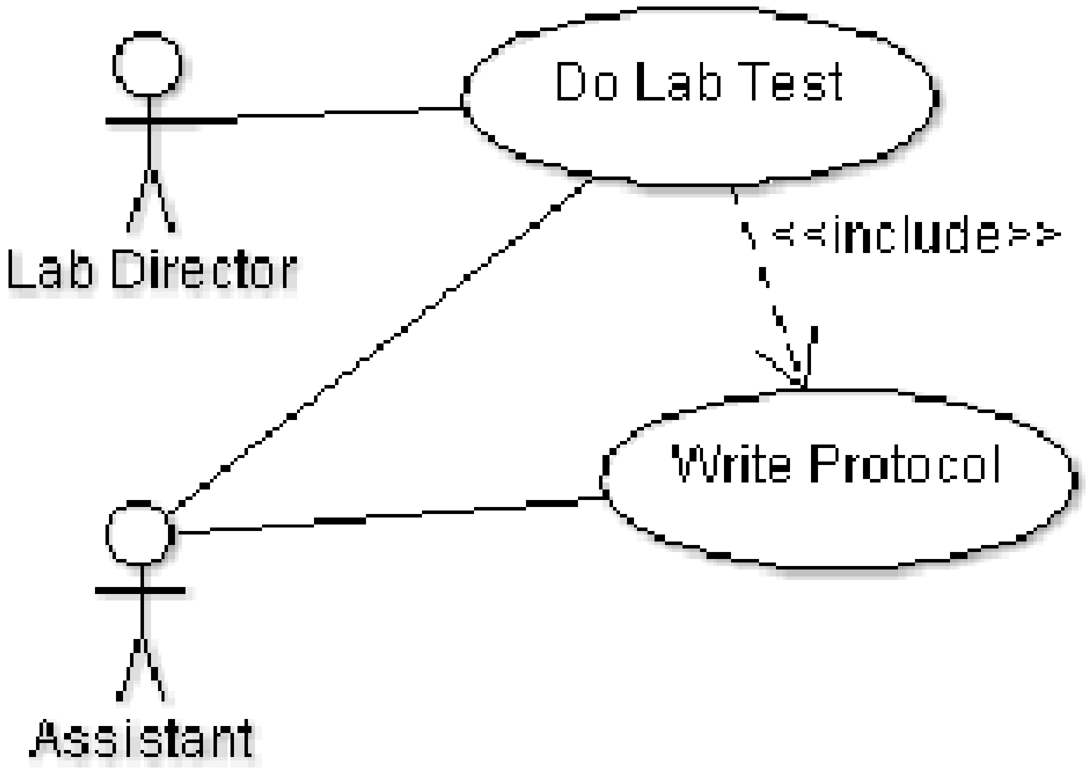
Answer : C

5. Consider the following scenario

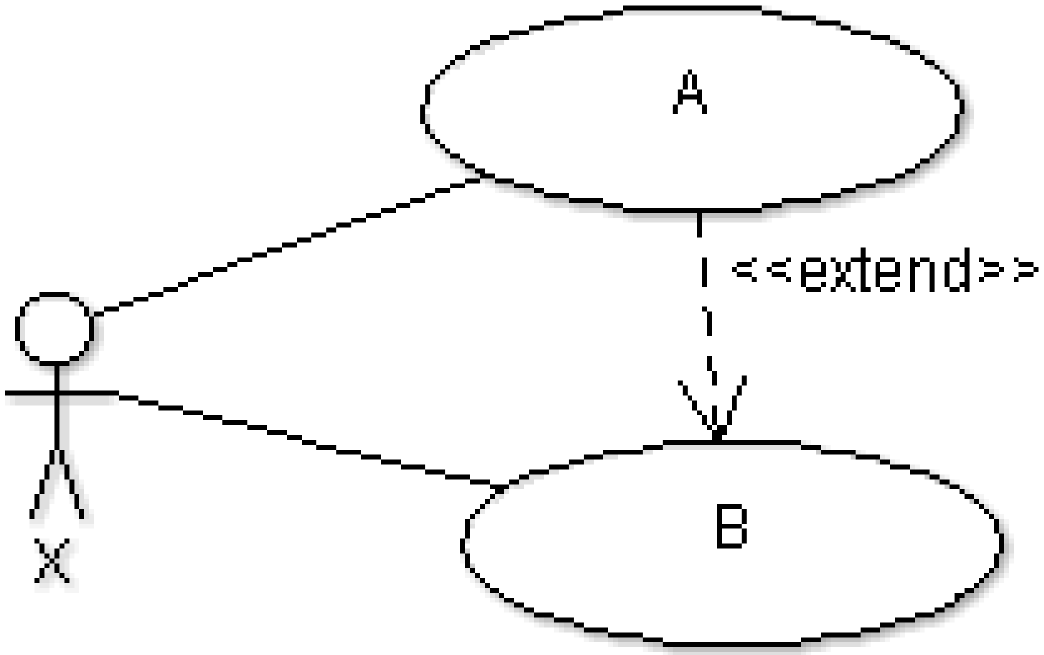
"The lab director does a lab test together with his assistant. The assistant always has to write a protocol during the lab test."

- Draw a Use Case model for the above situation.

5. Answer



6. The <<extend>> relationship in the following diagram means:



6. The <<extend>> relationship in the following diagram means:

- (a) B might or might not invoke A
- (b) A might or might not invoke B
- (c) B is optional behavior to A
- (d) B always has to invoke A.
- (e) A cannot be executed without B

Answer : A